NAICHENG JIANG

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RESEARCH INTERESTS

3D generative modelling and spatial representations; vision-language understanding; reliable multimodal AI, including applications in medical imaging.

EDUCATION

3+2 bachelor's–master's study pathway.

KTH Royal Institute of Technology, Stockholm, Sweden

MSc in Information and Network Engineering | Aug 2025 – Jun 2027 (expected)

Weighted grade average: **4.72/5.00** (self-calculated from completed graded courses).

Shanghai Jiao Tong University, Shanghai, China

BSc in Electrical and Computer Engineering | Aug 2022 – Jun 2026

UM-SJTU Joint Institute (Global College) | GPA: 3.57/4.00

HONORS AND AWARDS

Outstanding Graduate Award, Undergraduate Class of 2026 — Shanghai Jiao Tong University

Selected for the GIFT Future Scholars Program, Shanghai Jiao Tong University (2024)

Yu Liming Scholarship, UM-SJTU Joint Institute (2024)

RESEARCH EXPERIENCE

VecSetX — Compositional 3D Generation from Incomplete Supervision

Research collaboration | Feb 2026 – Present

Co-first author; manuscript in preparation.

Collaborative framework for conditional completion and unconditional generation of 31 anatomical structures, learning from four datasets with complementary, incomplete annotations.

- Extended and fine-tuned a multi-source 3D tokenizer, achieving 0.9229 mean reconstruction Dice on 204 validation cases at 256^3 resolution; delivered frozen latent representations for generative modelling.
- Implemented an experimental conditional diffusion transformer with direct query-to-context cross-attention and matched/shuffled-context controls; diagnosed context neglect in an earlier conditioning variant.
- Built source-holdout and cross-source generation comparisons against class-mean, regression and EM baselines, with paired case-bootstrap analysis; documented source-dependent structural gains.
- Identified systematic wall–cavity intrusion and examined geometric-rule sensitivity; distinguished consistency of jointly generated structures from patient-specific accuracy when target annotations were unavailable.

Pulse — Structural Reliability and Failure Diagnosis

Research Contributor — Reliability Analysis | 2026

- Analyzed 50 ATM22 airway cases through a reproducible evaluation of connectivity, component-level confidence and overlap metrics, localizing structural failure modes.
- Identified cases where largest-component filtering improved overlap scores while removing valid, high-confidence structures, motivating evaluation of corrections that preserve already-correct predictions.

CURRENT AND PLANNED PROJECTS

Flow Matching for Image Reconstruction | KTH EQ2443 | Ongoing

Course project at KTH. Developing the experimental plan for image reconstruction from perturbed originals, relating flow-estimation accuracy to reconstruction quality and computational cost.

3D Vision-Language Understanding | Planned research

Planned research connecting 3D representations with language-based understanding, with an interest in spatial reasoning and consistency between model responses and visual evidence.

TECHNICAL SKILLS

Python, MATLAB, C++; PyTorch, OpenCV, NumPy/SciPy; Linux, Git, LaTeX, Slurm, LUMI/HPC, CUDA.
Chinese (native); English (professional working proficiency).